

Prior Design Attempt #1 - Model Design Attempt 1

Source

In order to get a better understanding of what we should do for our project we used the University of Maryland burn day slide presentation. The presentation provided us with an explanation of the project, criteria, and strategies in order to make our model successful. We also researched videos on similar models and what materials they used to make them.

Image

I wasn't able to get a picture of this model

Details

The structure of the model was 18x12 and the ceiling height was 12. We had one doorway that connected with the two rooms that was 6.5 x 2.5. We had a window that was 3" wide and 2" high. The two beds were 6 x 3 x 2. The desk

Analysis

Since this design was the first model we didn't have many unsuccessful parts that didn't work since we weren't testing it yet

Prior Design Attempt #2 - Final design

To make our final design we used the kit provided by UMD for most of the materials we needed for our model. We had referred to the burn manual to make our furniture and walls up to scale. Since we needed more piping and something to hold our water we used Amazon to search for more material not provided in the kit.

Image

I don't have a picture of this model

Details

Our system was on the roof in order to stay in the building. The ceiling was removable. There were two bedrooms one was 6 x 12 and the other 12 x 12. The entire structure was 18 x 12. We were also provided with a glass wall. The doorway was 6.5" x 2.5" and window was 3" wide and 2" high.

Analysis

The part of our design that wasn't successful was the smoke detector. Our light and sound worked, but detecting the fire part didn't. According to the design requirements, we need to have light, sound, and the fire needs to be detected within 5 seconds which was the problem we had.